Questions

Q1.

Figure 12 is a diagram showing a lens, with some light rays passing through it.

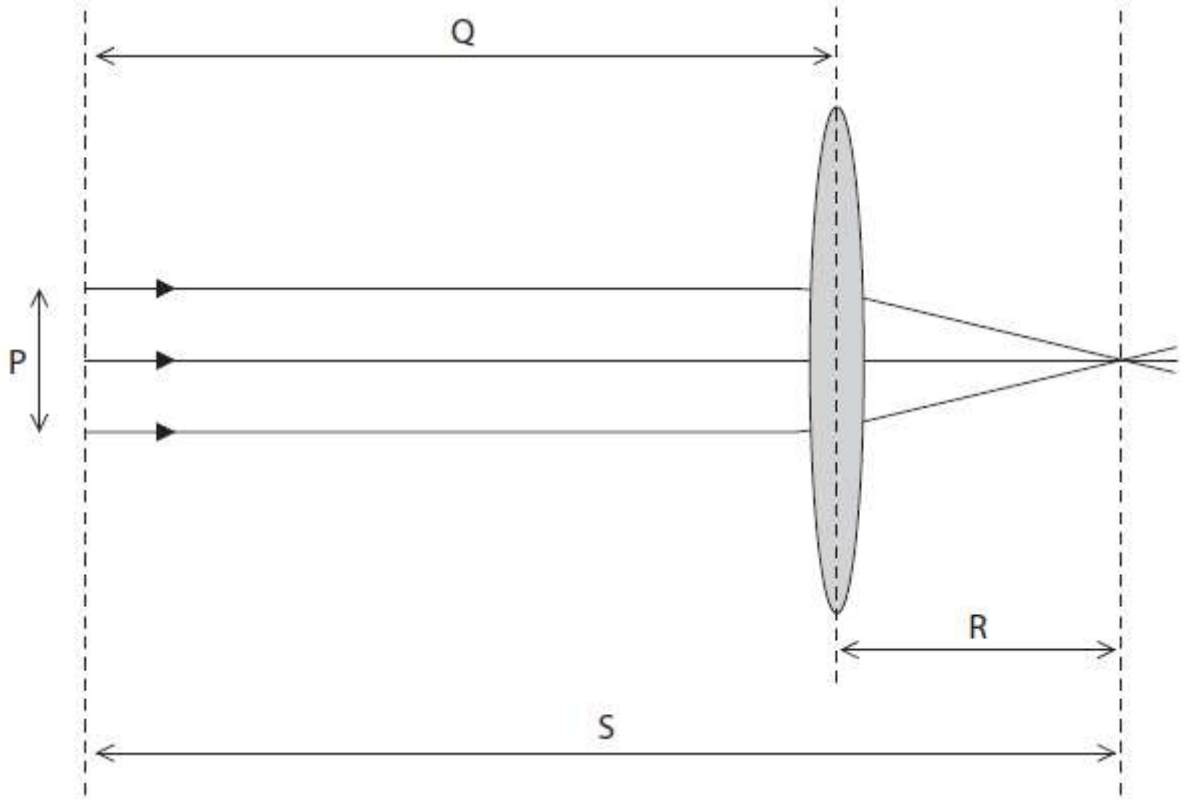

Figure 12

(i) This diagram shows a

(1)

- ☐ **A** converging lens forming a real image
- ☐ **B** diverging lens forming a real image
- ☐ **C** converging lens forming a virtual image
- ☐ **D** diverging lens forming a virtual image

(ii) Which length, labelled on Figure 12, shows the focal length of the lens?

(1)

- ☐ **P**
- ☐ **Q**
- ☐ **R**
- ☐ **S**

(Total for question = 2 marks)

Q2.

A student looks through a blue filter at a green leaf on a white background.

Describe the colours that the student sees.

(2)

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.....

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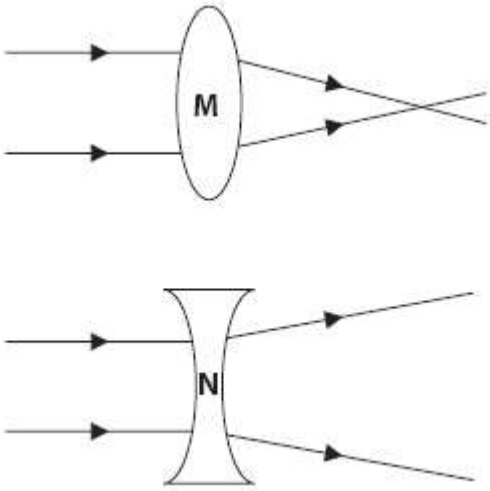
.....

(Total for question = 2 marks)

Q3.

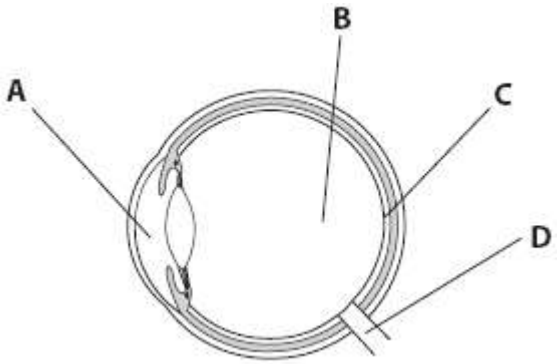
(a) The diagrams show rays of light passing through two lenses labelled **M** and **N**.



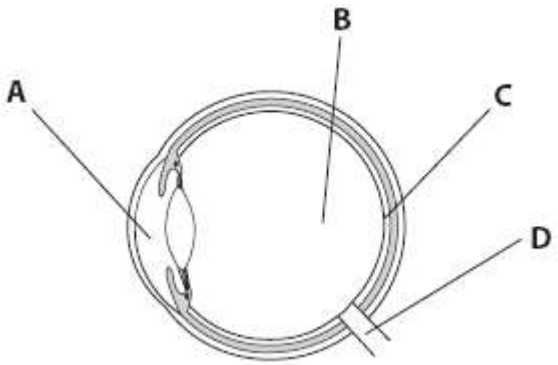
(i) Complete the sentence by putting a cross (■) in the box next to your answer.

The light changes direction as it leaves the lenses.
This is called

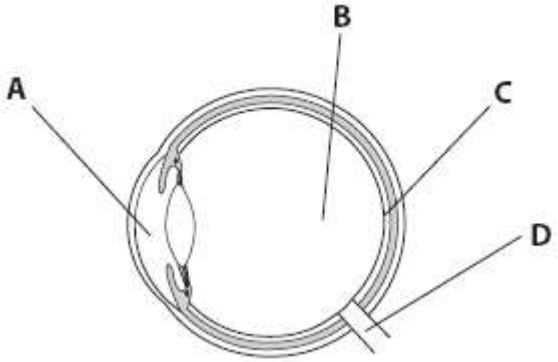
(1)



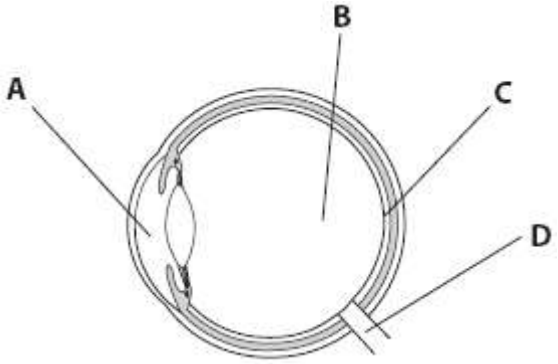
A contraction



B diffraction



C reflection



D refraction

(ii) The focal length of lens **M** is 2 m.

Calculate the power of lens **M**.

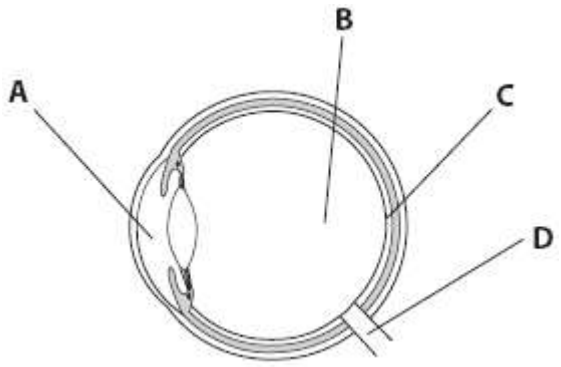
(2)

power = ...D

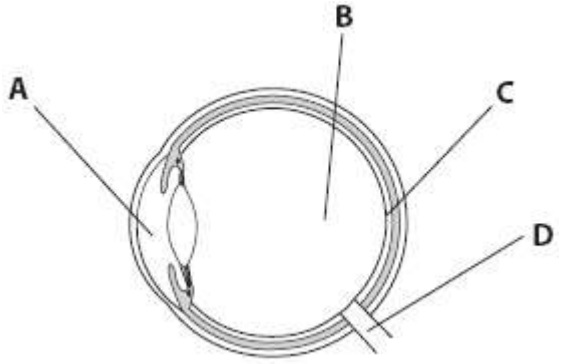
(b) The diagram shows an eye of a short sighted person.
A diverging lens is used to correct short sight.

(i) The eye will form an image of a distant object.
Which letter shows where the image will form for the short sighted eye?
Put a cross (■) in the box next to your answer.

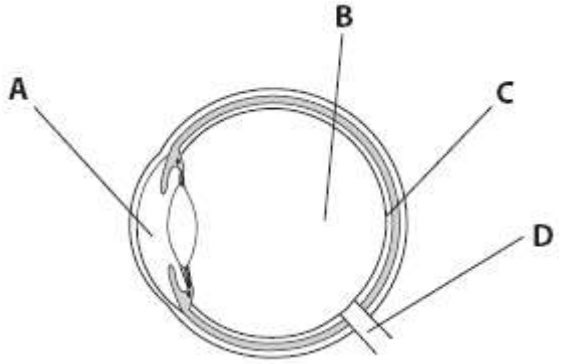
(1)



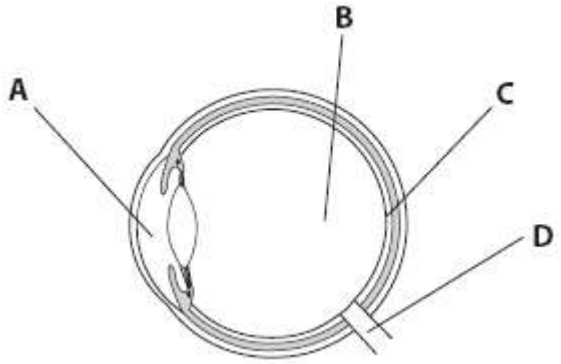
A



B



C



D

(ii) Draw on the diagram a diverging lens in a position which would correct the short sightedness.

(1)

(iii) Describe how the diverging lens corrects this defect of vision..

(2)

.....

.....

.....

.....

(iv) Suggest one way of correcting short sight other than wearing glasses.

(1)

.....

.....

(Total for Question = 8 marks)

Q4.

Diagram 1 shows an eye.

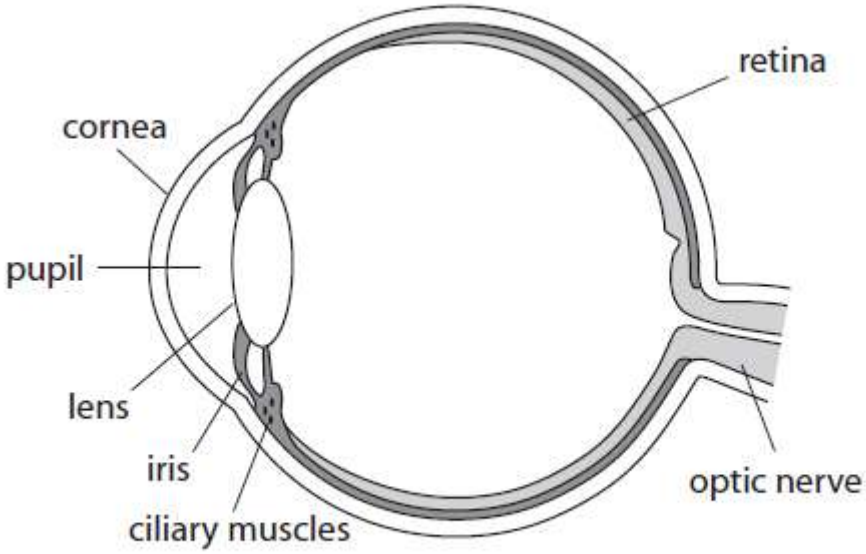


Diagram 1

(a) (i) Use words from the labels on diagram 1 to complete the sentence.

The parts of the eye which focus light are the

(2)

..... and the .....

(ii) Lenses of different powers are shown in diagram 2.

Diagram 2

Explain which lens has the greater power.

(2)

.....

.....

.....

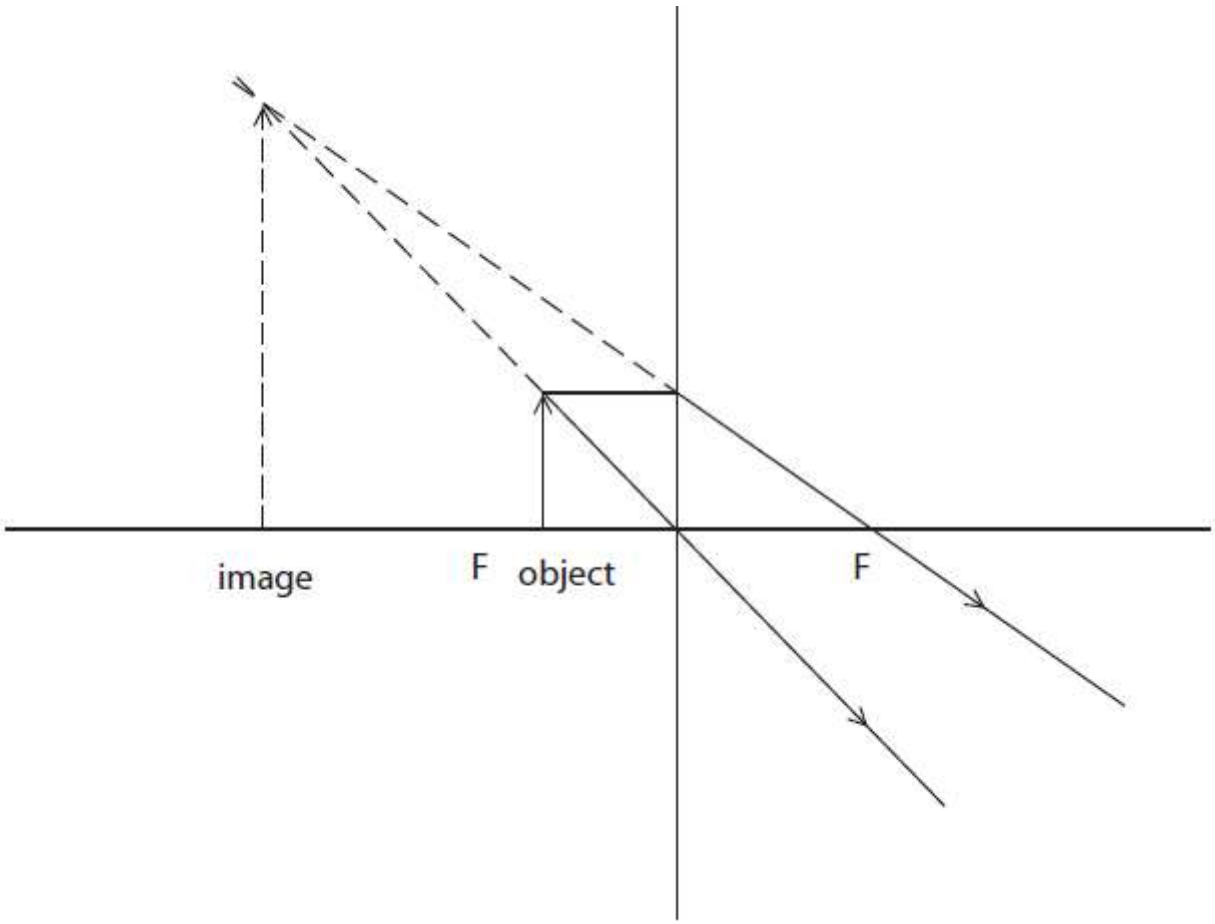
.....

(iii) Complete the sentence by putting a cross (☒) in the box next to your answer.
The unit for the power of a lens is the

(1)

☐ **A** decibel
☐ **B** dioptre
☐ **C** metre
☐ **D** watt

(b) The ray diagram shows the image of an object formed by a converging lens when it is used as a magnifying glass.



(i) The object has a height of 4 cm.
Estimate the height of the image.

(2)

height of image = ... cm

(ii) Which of these describes the image formed by this converging lens when it is used as a magnifying glass?

Put a cross (☒) in the box next to your answer.

(1)

☐ **A** inverted, real
☐ **B** right way up, real
☐ **C** inverted, virtual
☐ **D** right way up, virtual

(c) Explain why a converging lens is used for the treatment of long sight.

You may draw a diagram to help with your answer.

(2)

.....

.....

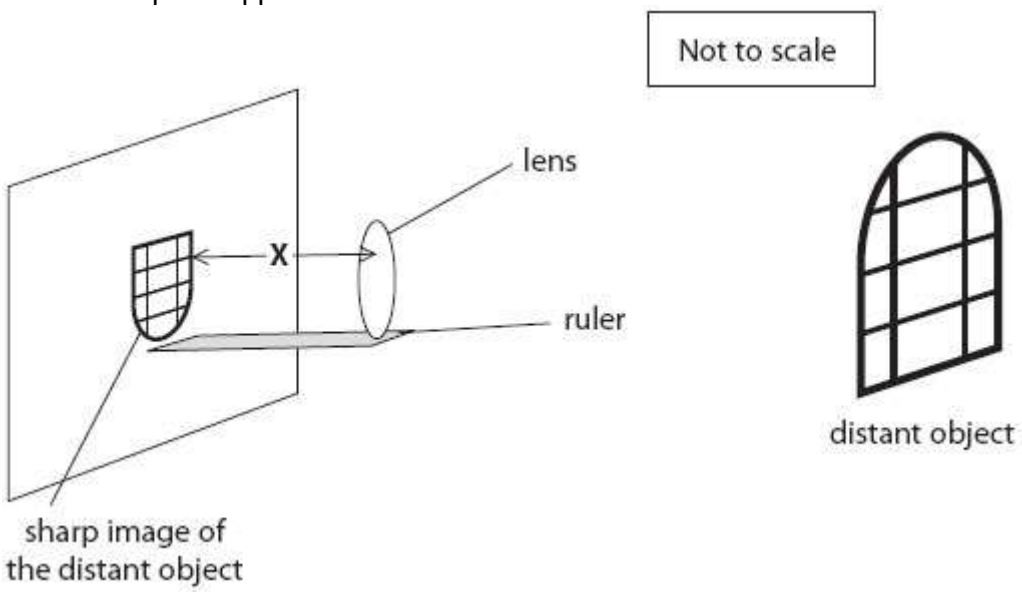
.....

.....

(Total for Question = 10 marks)

Q5.

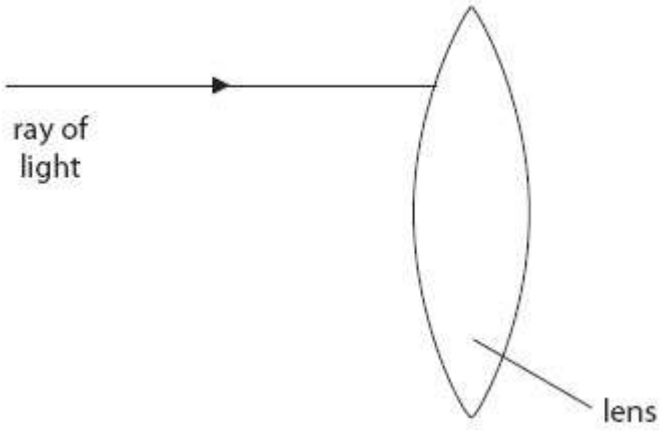
(a) Some students investigate a converging lens.
The students set up the apparatus as shown.



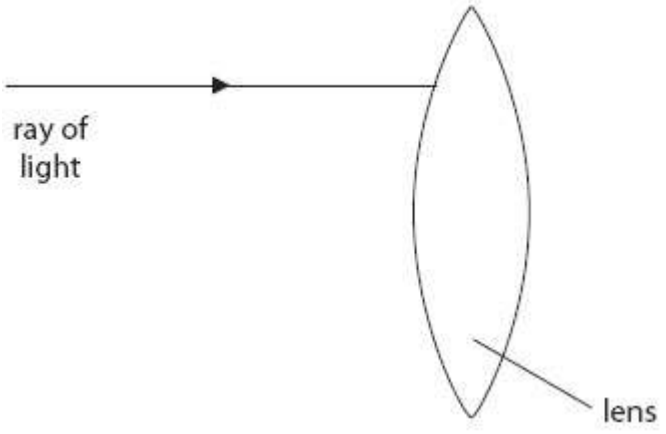
Complete the sentence by putting a cross (■) in the box next to your answer.

(i) The distance **X** is

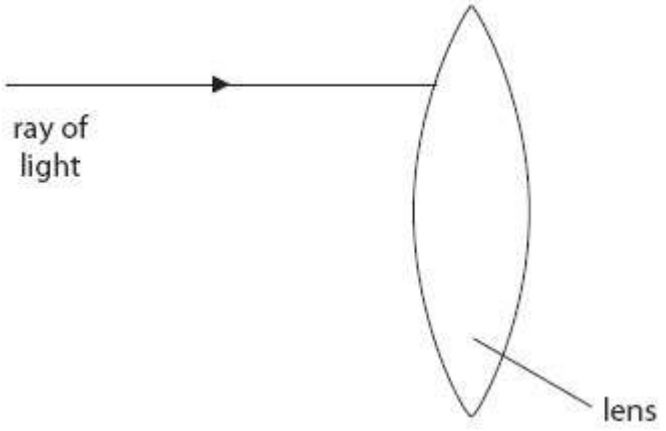
(1)



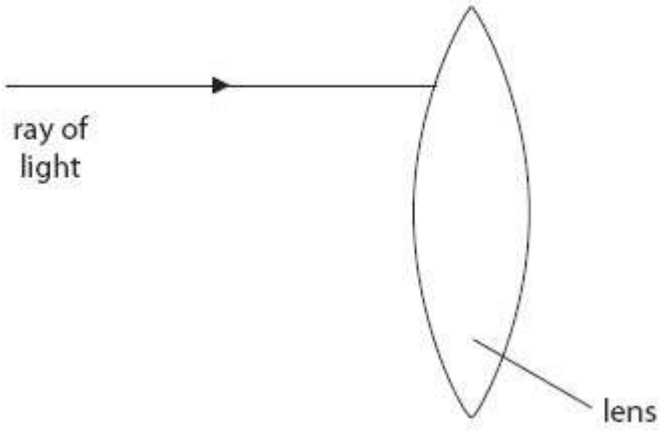
A the focal length



B the object distance



C the eyepiece distance



D the magnification

(ii) Use words from the box to complete the sentences.

(2)

bigger than	diverging	real
smaller than	the same as	virtual

The size of the image is .....the distant object.
The type of image formed on the screen is a .....image.

(b) The diagram shows a ray of light as it arrives at a lens.

Draw the path of the ray inside the lens.

(1)

(c) The students use a telescope to view the Moon.

Light from the Moon takes 1.3 s to reach the students.

The speed of light is 300 000 km/s.

Calculate the distance to the Moon.

(2)

distance = speed × time

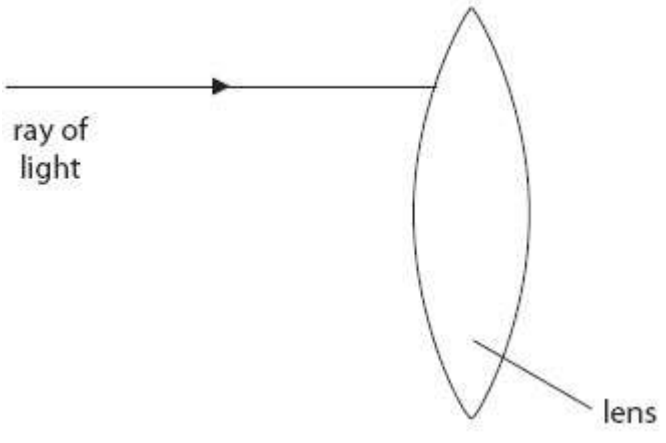
.....

(d) Complete the sentence by putting a cross (■) in the box next to your answer.

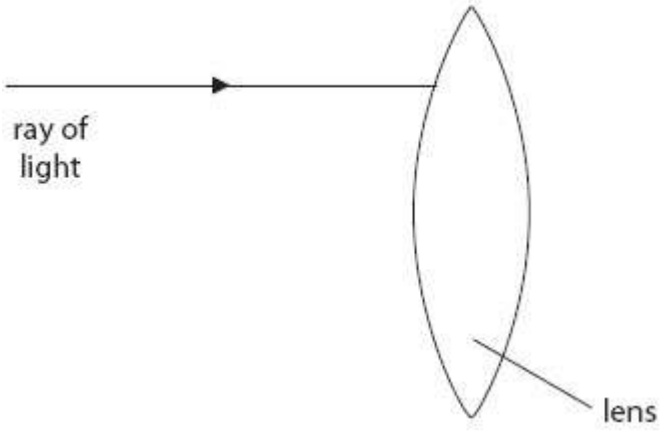
A satellite orbits the Moon.

Radio waves from this satellite transfer

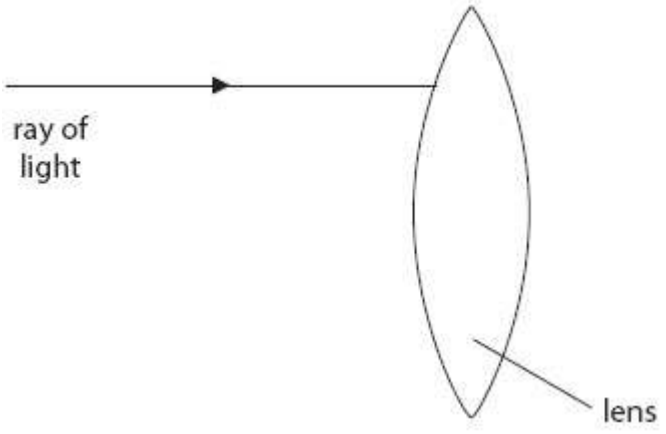
(1)



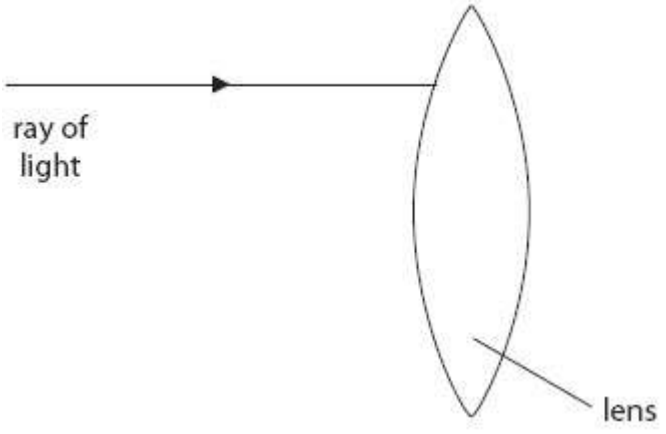
A matter only



B energy and matter



C information and matter



D energy and information

(Total for Question is 7 marks)

Q6.

(a) Put a cross (☒) in the box next to your answer.

Which of these is the biggest?

(1)

☐ **A** the Solar System

☐ **B** a galaxy

☐ **C** a nebula

☒ **D** the Universe

(b) These are four stages in the evolution of a star similar to the Sun.
They are **not** in the correct order.

1. main sequence star
2. white dwarf
3. red giant
4. nebula

Write down the stages in the correct order.

(2)

The first stage has been done for you.

..... nebula ..... .

.....

.....

.....

(c) (i) The chart shows the electromagnetic (EM) spectrum.
Some parts of the spectrum have been labelled.

radio	**P**	**Q**	visible light	**R**	**S**	gamma rays

State the name of part **Q**.

(1)

.....

(ii) Three telescopes using different parts of the spectrum have been added to the chart.

radio	**P**	**Q**	visible light	**R**	**S**	gamma rays
Jodrell Bank			Hubble			Compton

Jodrell Bank is located near Manchester, whereas Hubble and Compton are in space.

Explain why some telescopes are located outside the Earth's atmosphere.

(2)

.....

.....

*(d) Describe how modern telescopes have contributed to our understanding of the Universe.

(6)

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(Total for Question is 12 marks)

Q7.

Figure 5 is a graph showing the intensity–wavelength curves for two hot objects, L and M.

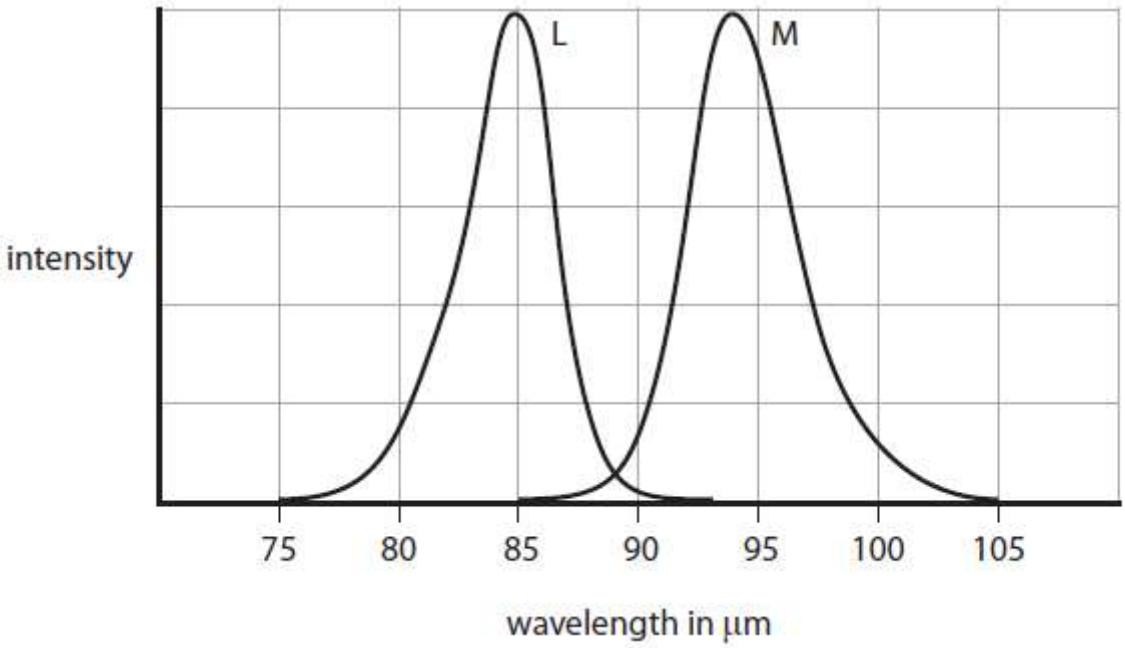


Figure 5

(i) Estimate the wavelength where the intensity is at a maximum for each of the objects.

(2)

wavelength at maximum intensity for object L = ... μm

wavelength at maximum intensity for object M = ... μm

(ii) State, with a reason, which object is the hotter object.

(1)

Object

.....

Reason.

.....

.....

(Total for question = 3 marks)

Q8.

Figure 9 shows an image taken at night using a camera that is sensitive to infrared radiation.

Figure 9

Radiation with shorter wavelengths shows as brighter areas in this image.

Explain why the people can be seen against the background in this image.

(2)

.....

.....

.....

.....

(Total for question = 2 marks)

Q9.

This question is about the electromagnetic spectrum.

X-rays and microwaves are both parts of the electromagnetic spectrum.

Use words from the box to complete the sentences.

(3)

amplitude	frequency	longitudinal	speed	transverse	wavelength

X-rays and microwaves are both ..... waves.

In a vacuum, x-rays and microwaves always have the same .....

X-rays always have a higher ..... than microwaves.

(Total for question = 3 marks)

Q10.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Ultraviolet light has a higher frequency than infrared light.

Which of these colours of visible light has the highest frequency?

(1)

☐ **A** blue
☐ **B** green
☐ **C** orange
☐ **D** yellow

(Total for question = 1 mark)

Mark Scheme

Q1.

Question Number	Answer	Mark
(i)	A converging lens forming a real image **The only correct answer is A** *B 'diverging lens forming a real image' is incorrect since the lens drawn in the diagram is not diverging* *C 'converging lens forming a virtual image' is incorrect, since the diagram shows rays coming to a focus (on a screen if placed there)* *D 'diverging lens forming a virtual image' is incorrect, since the diagram shows neither a diverging image or the formation of a virtual image*	**(1)** AO 3 1a

Question Number	Answer	Mark
(ii)	C **'R' The only correct answer is 'R'** ***'P'*** *is incorrect, being a distractor, showing a width of rays* ***'Q'*** *is incorrect, being a distractor showing an arbitrary distance to the lens distance* ***'S'*** *is incorrect, being a distractor showing an overall distance*	**(1)** AO 3 1b

Q2.

Question number	Answer	Mark
	An answer that provides a description by making reference to: leaf appears black (1) background appears blue (1)	**(2)**

Q3.

	Answer	Acceptable answers	Mark
(a)(i)	☒ **D** refraction		**(1)**
(a)(ii)	substitution: (1)	Award 2 marks	**(2)**

	power = 1/ 2 evaluation: (1) 0.5 (D)	for correct answer no working	
(b)(i)	☒ **B**		**(1)**
(b)(ii)	in front of eye (1)	any other diverging lens (lens which is narrower at the centre than at the ends) Reject lens inside the eye	**(1)**
(b)(iii)	a description including any **two** from: • to spread out the light/ rays (1) • to produce focused image (1) • (focused image) on the retina (1)	make image distance longer sharp/clear image back of eye	**(2)**
(b)(iv)	contact lenses (1)	laser surgery/ eye lens replacement	**(1)**

Q4.

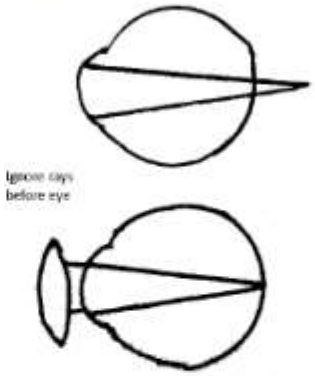
Question Number	Answer	Acceptable answers	Mark
(a)(i)	• cornea (1) • lens / ciliary muscles (1)	In either order accept misspellings where meaning is clear	**(2)**

Question Number	Answer	Acceptable answers	Mark
(a)(ii)	An explanation linking: • Lens B (1) with one of • smaller radius of curvature • (because) it is thicker (1) • (because) bends/refracts light more (1)	 more curved fatter / wider (not 'bigger') has a shorter focal length	**(2)**

Question Number	Answer	Acceptable answers	Mark
(a)(iii)	B dioptre		**(1)**

Question Number	Answer	Acceptable answers	Mark
(b)(i)	• 4 x3 (1) • Evaluation 12(cm) (1)	10.0 (cm) to 15.0(cm) Correct answer with no working for 2 marks	**(2)**

Question Number	Answer	Acceptable answers	Mark
(b)(ii)	D right way up, virtual		**(1)**

Question Number	Answer	Acceptable answers	Mark
(c)	An explanation linking any two of the following: • the eyeball is too short / the image is formed behind the retina (eye) / the eye lens does not bend the light enough / is not powerful enough / person cannot see objects close up (clearly) (1) • (lens) bends / refracts the light more (1) • (lens) focuses the image (of an object) on the retina / back of eye (1)	Credit diagrams  reject contradictions where short sight is shown or described i.e. lens focusing rays before the retina.	(2)

(Total for Question = 10 marks)

Q5.

	Answer	Acceptable answers	Mark
(a)(i)	A the focal length (1)		**(1)**
(a)(ii)	smaller than (1) real (1)		**(2)**
(b)	• Any (more or less) straight ray which changes direction inside the lens (1)	Ray does not need to touch far side. Allow slight discontinuities Ignore any ray drawn beyond the 2nd surface and any reflected ray(s). Ignore any extra incident rays.	**(1)**
(c)	substitution into given equation (1) 1.3 × 300 000 evaluation (1) 390 000 (km)	Power of 10 error max 1 mark 3.9×10^5 (km) 2 marks for correct numerical answer with no working shown Ignore any unit given by candidate.	**(2)**
(d)	D energy and information (1)		**(1)**

Total for Question = 7 marks

Q6.

	Answer	Acceptable answers	Mark
(a)	D the Universe (1)		**(1)**
(b)	(nebula) main sequence (star) (1) AND red giant white dwarf (1) All three in correct order for 2 marks	Red Giant White Dwarf (Main sequence) (1)	**(2)**
(c)i	infrared (radiation)/(rays) (1)		**(1)**
(c)ii	An explanation linking any two from • above the clouds / no clouds/ no weather (1) • image is clearer/more detailed/ not distorted/not blurred (1) • no light pollution (1) • (some) telescopes use gamma/ X-rays/ultraviolet /infrared/microwaves (1) • no absorption (by atmosphere) of gamma/ Xrays/ ultraviolet /infrared/ microwaves (1)	Credit to be given for stating that all telescopes would be better in space, but size and weight may exclude e.g. Jodrell Bank from space. no air/dust/pollution wider field of view/ can use anytime IG NORE 'see further' IGNORE 'it is closer (to the stars/planets)' IGNORE: references to improving understanding / knowledge of space	**(2)**

		Indicative Content	Mark
QWC	***(d)**	A description including some of the following points • improved QUALITY eg higher or better magnification/ detail/resolution or clearer/brighter image **OR** MORE INFORMATION (than with naked	**(6)** **Exp**

		eye) of image/data eg new planets/stars/nebulae/pulsars (This could be extra detail for greater magnification/resolution only) • detection of (non-visible) electromagnetic WAVES eg Xray / UV / IR/ radio • TECHNOLOGY that enable collection of more data eg reflecting telescope/arrays ***and/or*** additions eg computer-aided /photographic connections or larger (objective) lens/mirror • POSITION of telescopes – eg orbital/outside atmosphere/on top of mountains/away from atmosphere/rays not absorbed/obscured/scattered by atmosphere. Ignore 'Hubble' or 'Compton'.	
Level	**0**	No rewardable content	
1	**1 - 2**	• a limited description e.g. mention of any one example such as "magnifies stars/planets" OR "discovering new planets/stars" • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy	
2	**3 - 4**	• a simple description e.g. mention of either two of the improvements **OR** extra detail about one of the improvements eg improvement **plus** example (ie Magnifies planets *so that craters/mountains may be seen)* • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately	

		• spelling, punctuation and grammar are used with some accuracy
3	**5 - 6**	• a detailed description e.g. mention of three (or more) improvements **OR** two improvements **plus extra detail about one of them (ie Telescopes in space can detect XRays *that would be absorbed by the atmosphere*)** • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors

(Suitable extra detail shown in italics in examples above)

Total for Question = 12 marks

Q7.

Question Number	Answer	Additional guidance	Mark
(i)	wavelength for object L = 84.(0) to 85(.0) (μm) (1) wavelength for object M = 93.(0) to 95.(0)(μm) (1)	estimates, so we are being generous with this	**(2)** AO 3 2a

Question Number	Answer	Additional guidance	Mark
(ii)	object L and shorter wavelength (1)		**(1)** AO 3 1a

Q8.

Question number	Answer	Additional guidance	Mark
	An explanation that combines application of knowledge (1 mark) and reasoning (1 mark) linking: (faces of) people are at a higher temperature than the background (1) therefore they emit more (infrared) at shorter wavelengths than background (1)	accept higher frequency / higher intensity	**(2)**

Q9.

Question number	Answer	Additional guidance	Mark
	transverse (1) speed (1) frequency (1)	must be in correct order. allow any recognisable spelling.	**3** **AO1.1**

Q10.

Question number	Answer	Mark
	☒ A blue Green, orange and yellow all have a lower frequency than blue	**(1)**